

NAME:TANKISO MASOEBE

OPERATING SYSTEM(ADVANCED)WK7

1.SCENARIO ANALYSIS:

Security Setup for a Cloud Operating System

A cloud operating system requires a strong security setup to protect data, applications, and users from unauthorized access and cyber threats. In this scenario, an organization uses a cloud-based operating system to manage its business applications and store important files. The system uses **multi-factor authentication (MFA)** so users must provide a password and a verification code before accessing their accounts. **Role-based access control (RBAC)** is also implemented to ensure employees can only access resources required for their jobs. Data is protected using **encryption** both when stored and when transmitted across networks. A firewall monitors network traffic and blocks suspicious connections. The organization also installs security updates regularly to fix vulnerabilities. **Antivirus and intrusion detection systems** monitor the environment for malicious activity. Finally, regular backups are performed so important data can be restored after accidental deletion, system failure, or a cyberattack. This security setup improves confidentiality, integrity, and availability.

2.CONCEPT RESEARCH:

Role-Based Access Control (RBAC)

Role-Based Access Control (RBAC) is a security method that manages access to computer systems based on a user's job role. Instead of giving each user individual permissions, administrators assign permissions to roles such as manager, administrator, lecturer, or student. Users are then assigned one or more appropriate roles. RBAC helps organizations control access to sensitive information and reduce unauthorized activities. For example, a system administrator may have permission to manage users, while a student may only view learning materials. RBAC also makes access management easier because permissions can be changed by updating roles rather than individual accounts. This improves security and simplifies administration.

3.TOOL PRACTICE:

Encryption in VirtualBox – Reflection

I configured encryption for a virtual machine in VirtualBox to protect the data stored within the virtual machine. The encryption feature helps prevent unauthorized users from accessing the virtual machine's virtual disk and sensitive information. During the practice, I enabled encryption in the virtual machine settings and created an encryption password. This showed me how encryption can protect data even if someone gains

access to the virtual machine files. Encryption is important because it provides confidentiality and reduces the risk of data theft. However, the encryption password must be kept secure because losing it may prevent access to the encrypted virtual machine. Overall, this practice improved my understanding of data protection.

4.APPLICATION PRACTICE:

Security Policy

A security policy is a set of rules and guidelines designed to protect an organization's computers, networks, applications, and information. The policy requires users to create strong passwords, use multi-factor authentication, and avoid sharing login credentials. Employees must only access information necessary for their job responsibilities. Antivirus software, firewalls, and regular security updates should be used to protect systems from malware and cyberattacks. Important data must be backed up regularly and stored securely. Employees should report suspicious emails, unauthorized access, or security incidents immediately. The purpose of the security policy is to reduce security risks, protect sensitive information, prevent unauthorized access, and ensure that users understand their responsibilities when using organizational IT resources.